

PORI TAHKOLUOTO HARBOUR, Finland

WMO: 027510

Lat: 61.630N

Lon: 21.376E

Elev: 15

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	5.6	-3.4	4.6	2.0	1.9	6.0	7.0	43.0	33.2	39.4	33.1	9.6	110	0.808

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	74.4	67.2	71.3	65.3	68.7	64.0	69.1	72.9	66.6	70.0	64.6	68.0	10.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	100.6	71.7	65.2	93.1	69.4	62.9	85.9	67.1	33.2	73.1	31.2	70.0	29.7	68.0	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.1	31.7	28.7	-2.5	79.5	6.7	4.6	-7.3	82.8	-11.3	85.5	-15.1	88.1	-20.0	91.4		
			-1.6	69.3	7.9	3.0	-7.3	71.4	-11.9	73.2	-16.4	74.8	-22.1	77.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.9	24.9	26.2	30.1	37.2	47.9	54.8	62.1	61.6	55.3	44.5	37.7	31.7
	DBStd	14.47	9.23	9.37	5.99	4.52	6.54	5.40	5.56	4.37	4.58	5.26	6.74	8.89
	HDD50	3711	778	668	618	386	120	12	0	0	13	180	369	567
	HDD65	8117	1243	1088	1083	833	532	311	127	123	290	636	819	1032
	CDD50	1128	0	0	0	3	54	156	374	360	173	8	0	0
	CDD65	59	0	0	0	0	1	5	36	17	0	0	0	0
	CDH74	106	0	0	0	0	4	9	54	39	0	0	0	0
	CDH80	9	0	0	0	0	0	1	5	3	0	0	0	0
Wind	WSAvg	14.6	15.6	15.5	15.0	12.8	12.9	12.9	12.2	13.8	14.3	15.7	16.4	18.6
Precipitation	PrecAvg	24.8	2.0	1.5	1.3	1.3	1.5	2.2	2.7	2.8	2.3	2.7	2.4	2.3
	PrecMax	32.2	3.9	3.1	2.3	2.4	3.3	4.3	5.3	5.4	5.0	6.3	5.5	4.3
	PrecMin	20.2	0.3	0.1	0.2	0.0	0.2	0.7	0.1	1.0	0.5	0.3	0.4	0.3
	PrecStd	3.0	0.9	0.8	0.5	0.6	0.8	0.9	1.3	1.3	1.4	1.4	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	38.5	43.0	58.9	72.4	74.5	78.7	78.9	68.2	56.6	50.0	44.8
		MCWB	37.8	36.5	37.3	48.2	57.4	62.6	69.4	69.4	62.3	53.7	48.2	43.3
	2%	DB	38.1	37.0	39.2	49.2	64.4	69.4	75.0	73.1	64.0	54.2	48.2	42.9
		MCWB	36.8	35.5	36.9	43.6	57.8	62.6	68.3	67.1	60.1	51.6	47.2	41.1
	5%	DB	36.9	36.0	37.7	44.6	60.8	65.5	72.3	70.1	62.5	52.7	46.4	41.3
		MCWB	35.7	34.9	35.9	41.1	56.1	60.6	67.1	64.9	59.3	50.3	44.9	39.5
	10%	DB	35.5	35.0	36.6	42.3	56.9	62.0	69.7	67.0	61.0	51.0	45.0	40.1
		MCWB	34.6	34.2	35.1	39.8	52.2	58.0	65.6	62.7	58.1	48.8	43.6	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	36.9	38.8	50.2	60.9	65.9	71.7	71.6	62.8	54.3	48.6	43.9
		MCDB	38.9	37.8	41.0	57.4	66.6	70.9	76.9	76.3	66.8	55.9	49.4	44.6
	2%	WB	37.2	35.9	37.2	44.6	58.6	63.1	70.0	68.1	61.0	52.3	47.4	41.4
		MCDB	37.7	36.5	38.6	48.8	64.3	68.1	73.9	72.0	63.4	53.8	48.2	42.5
	5%	WB	35.8	35.0	36.3	41.8	56.0	60.9	68.0	65.6	59.7	50.8	45.2	39.8
		MCDB	36.6	35.6	37.5	43.8	61.0	65.3	71.3	69.2	61.8	52.1	46.1	40.9
	10%	WB	34.6	34.3	35.4	40.1	52.3	58.4	66.0	63.7	58.6	49.2	43.8	38.8
		MCDB	35.4	34.9	36.5	41.9	56.2	61.6	69.1	66.4	60.5	50.5	44.7	39.8
Mean Daily Temperature Range	5% DB	MDBR	7.3	6.5	7.0	6.0	7.0	6.4	6.4	5.9	5.7	5.2	4.7	5.5
		MCDBR	4.8	4.4	5.9	10.3	12.2	12.1	10.3	9.4	7.6	4.9	4.6	4.9
		MCWBR	4.4	3.7	4.7	6.6	7.0	6.7	6.0	6.0	5.2	4.8	4.5	5.0
	5% WB	MCDBR	4.5	4.3	5.7	9.1	11.7	11.0	8.4	7.4	6.3	4.7	4.2	4.7
		MCWBR	4.2	3.9	4.8	6.1	7.2	6.9	5.6	5.6	5.0	4.8	4.3	4.8
Clear-Sky Solar Irradiance	taub	0.264	0.308	0.326	0.366	0.364	0.356	0.382	0.369	0.344	0.327	0.283	0.350	
	taud	2.137	2.239	2.313	2.329	2.420	2.479	2.431	2.456	2.506	2.394	2.186	2.519	
	Ebn at Noon	138	207	250	261	272	277	265	259	245	201	128	88	
	Edh at Noon	11	21	27	32	32	31	32	29	23	18	11	7	
All-Sky Solar Radiation	RadAvg	58	233	720	1245	1761	1937	1778	1383	818	334	85	35	
	RadStd	8	45	82	126	146	113	151	120	82	56	15	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page